

Below is a professional **Technical Requirements Document (TRD)** for your **AI Recipe Generator** project. You can give this directly to Claude, ChatGPT, or your development team.

# Technical Requirements Document (TRD)

## Project Name

**AI Recipe Generator**

## Version

1.0

## Technology Stack

### Frontend

- Flutter 3.x
- Dart
- Material 3

### Backend

- Firebase

### AI

- Google Gemini API

## Database

- Cloud Firestore

## Authentication

- Firebase Authentication

## Storage

- Firebase Storage

## State Management

- Provider

## IDE

- VS Code

## Version Control

- Git
- GitHub

# Theme

Primary Color

#8B5E3C

Background

#F6F2EE

Cards

White

Border Radius

16-25 px

Buttons

Rounded

Brown

White Text

## Fonts

Google Fonts

Preferred

Poppins

Fallback

Roboto

## Assets

assets/

images/

icons/  
animations/

# Screens

## Splash Screen

Requirements

Display Logo

Display App Name

Animation

Navigate automatically

## Login Screen

Components

Logo

Welcome Text

Subtitle

Email Field

Password Field

Remember Me

Forgot Password

Google Sign In

Sign Up

Validation

Responsive

## Register Screen

Fields

Name

Email

Password

Confirm Password

Google Sign Up

Already have account

Validation

## Forgot Password

Email

Reset Button

Snackbar

Navigate Back

## Home Screen

Greeting

Search Bar

AI Assistant

Categories

Popular Recipes

Trending Recipes

Recommended

Bottom Navigation

## Recipe Details

Image

Recipe Name

Cooking Time

Calories

Difficulty

Ingredients

Instructions

Nutrition

Favorite Button

Share Button

## Favorites

Saved Recipes

Delete

Search

Filter

## Profile

Profile Picture

Name

Email

Saved Recipes

Logout

Settings

## AI Module

User Input

Create healthy dinner under 500 calories.

Gemini Response

Recipe Name

Ingredients

Instructions

Cooking Time

Calories

Tips

## AI Chat

Chat Screen

User Messages

Bot Messages

Typing Indicator

Auto Scroll

Markdown Support

## Firestore

Collections

Users

Recipes

Favorites

Generated Recipes

Chat History



# User Collection

uid

name

email

photoUrl

createdAt

# Recipe Collection

recipeId

title

ingredients

instructions

calories

time

image

createdBy

timestamp

# Favorites Collection

userId

recipeId

## Authentication

Email Login

Register

Google Login

Forgot Password

Logout

Remember Login

## Validation

Email

Password

Confirm Password

Name

No Empty Fields

# Provider Classes

AuthProvider

RecipeProvider

# Services

AuthService

FirestoreService

GeminiService

# Navigation

Splash



Login



Home



Recipe



Profile



Favorites



Chat

## Error Handling

Network Error

Firebase Error

Gemini Error

Invalid Input

Empty Response

## Performance

Lazy Loading

Cached Images

Optimized Widgets

No Overflow

Responsive UI

## Responsive Support

Android Phones

Small Screens

Medium Screens

Large Screens

Tablets

## Security

Firebase Rules

Secure API Key

Authentication Required

Input Validation

## Git Workflow

main

develop

feature/login

feature/register

feature/home

feature/chat

Commit Format

feat: login UI completed

fix: textfield alignment

refactor: authentication service

docs: updated README

# Milestones

## Milestone 1

- Splash
- Login
- Register
- Forgot Password

## Milestone 2

- Firebase Authentication

## Milestone 3

- Home Screen

## Milestone 4

- AI Recipe Generator

## Milestone 5

- Recipe Details

## Milestone 6

- Favorites

## **Milestone 7**

- Profile

## **Milestone 8**

- AI Chat

## **Milestone 9**

- Final Testing

# **Coding Standards**

- Use reusable widgets.
- Use Provider for state management.
- Avoid duplicate code.
- Follow Clean Architecture.
- Use meaningful file and variable names.
- Keep UI responsive.
- Handle all loading and error states.
- Comment complex logic where necessary.